

# Intelligence as a Moving Target: Plenty of Room at the Bottom, Middle, and Beyond

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## Abstract

This essay posits that intelligence is not a static attribute but a dynamic, recursive property shaped by physical, cognitive, and cultural scales. Drawing upon the Relativistic ScalarVector Plenum (RSVP) framework, Active Inference, and the 4E approach to cognitive science (Embodied, Embedded, Enactive, Extended), it argues that human intelligence expands super-exponentially through multiscale interactions. Intelligence manifests as the entropic alignment of autonomous systems with evolving environments, countering static conceptions of artificial general intelligence (AGI) advanced by critics such as Gary Marcus. By integrating enaction, affordances, and phenomenology, the analysis demonstrates that education and collective recursion drive continuous reconfiguration, ensuring perpetual room for advancement at foundational, intermediate, and higher levels.

## 1. Introduction: The Fallacy of Static Intelligence

Debates surrounding artificial general intelligence frequently assume a fixed benchmark, often equated with “human-level” capabilities. Gary Marcus (2025) contends that large language models (LLMs) fail to generalize beyond training distributions, suggesting inherent limitations in achieving AGI. This perspective, however, presupposes intelligence as an invariant standard against which systems are measured.

In contrast, the Relativistic ScalarVector Plenum (RSVP) framework conceptualizes intelligence as a trajectory within a continuous field of potential ( $\Phi$ ), vector flows ( $\mathbf{v}$ ), and entropy ( $S$ ). Human cognition does not represent a ceiling but an initial phase in an evolving process. Intelligence emerges from recursive reorganization across scales, rendering it a moving target perpetually redefined by technological, educational, and cultural advancements.

## 2. Why Intelligence Is a Moving Target

Intelligence exhibits malleability, influenced by genetic, pharmacological, psychological, behavioral, and environmental factors. It may increase or decrease over time, defying rigid quantification.

Cultural and social contexts further underscore this dynamism. Definitions of intelligence vary across societies: some emphasize wisdom accrued with age, while others prioritize innovative capacities associated with youth. Societal norms shape identification and valuation, adapting to evolving values.

As technology progresses, understandings of intelligence transform. In the domain of artificial systems, ongoing debates reevaluate what constitutes intelligent behavior, illustrating the fluid nature of the concept.

## 3. Implications of Viewing Intelligence as a Moving Target

Recognizing intelligence's fluidity informs technological development, particularly in AI. Rather than pursuing direct replication of human traits, strategies should emphasize nuanced, adaptive goals that accommodate evolving definitions.

In education, this perspective fosters inclusive approaches. By acknowledging malleability, systems can nurture diverse aptitudes, enabling individuals to realize potential without confinement to static standards.

Broadly, this view dismantles myths surrounding intelligence, promoting an inclusive appreciation of human capabilities and reducing associated stigmas.

## 4. Education and the Thermodynamics of Generalization

Human intelligence operates as a collective endeavor. Educational structures, languages, and technologies form recursive networks that minimize entropy and sustain informational order. Institutions function as low-entropy attractors, facilitating knowledge transmission.

Let  $I(t)$  represent civilization's effective intelligence, with  $B(t)$  denoting physical capacity (compute per joule),  $M(t)$  heuristic efficiency (sparsity, causal inference), and  $T(t)$  educational transmissivity. The dynamics are captured by:

$$\frac{dI}{dt} = \alpha B(t)M(t)T(t)I(t)^\gamma, \quad \gamma \geq 1. \quad (1)$$

For  $\gamma = 1$ , growth is exponential; for  $\gamma > 1$ , it becomes super-exponential, reflecting self-

reinforcing amplification. Historical transitions such as the advent of writing or computing have elevated  $\gamma$  beyond unity, propelling sustained advancement.

## 5. Plenty of Room at the Bottom, Middle, and Beyond

Richard Feynman’s assertion of “plenty of room at the bottom” extends to cognition, where quantum and physical substrates unlock denser informational processing. Each substrate innovation deepens the scalar potential  $\Phi$ , enabling lower-entropy operations.

Equally, the “middle” layer encompassing sparse estimation, fuzzy logic, and probabilistic causal graphs compresses inference spaces, bridging neural and symbolic paradigms.

At higher levels, educational recursion renegotiates intelligence boundaries, reconfiguring couplings between  $\Phi$ ,  $\mathbf{v}$ , and  $S$ . These layers ensure intelligence evolves as field topology adapts.

## 6. Enaction and the Active Plenum

Insights from Julian Kiverstein and Darius (2024) in *Active Inference Insights 011: Affordances, Enaction, Phenomenology* frame cognition as enactive autonomy: systems maintain organization through environmental regulation via Markov blankets that couple rather than isolate.

RSVP extends this cosmologically. The plenum’s fields  $\Phi$  (potential),  $\mathbf{v}$  (flows),  $S$  (entropy) form a manifold where agents stabilize gradients through recursive interactions. Affordances emerge as stable alignments where  $\mathbf{v} \cdot \nabla \Phi \approx 0$ , extending cognition beyond biological boundaries to include tools and cultural artifacts.

This enactive view aligns with extended cognition, where external resources simplify inference by minimizing expected free energy.

### 6.1 Mathematical Interpretation of Enaction

Active Inference formalizes cognition as the minimization of *expected free energy*  $G$ , which measures the divergence between predicted and actual states of the world under a generative model:

$$G = \mathbb{E}_{q(s,\pi)} [D_{\text{KL}}(q(s, \pi) \parallel p(s, \pi)) - \log p(o|s, \pi)], \quad (2)$$

where  $s$  are latent states,  $\pi$  denotes policies (action trajectories), and  $o$  represents sensory outcomes. Minimizing  $G$  corresponds to selecting actions that reduce both uncertainty (epistemic drive) and expected surprise (pragmatic value).

In RSVP, the same principle appears in continuous form through the scalar–vector–entropy coupling. Let  $\Phi(x, t)$  represent informational potential,  $\mathbf{v}(x, t)$  the flow of activity (policy field),

and  $S(x, t)$  the local entropy density. Define the *field free energy functional*:

$$\mathcal{F}[\Phi, \mathbf{v}, S] = \int_{\Omega} \left[ \frac{1}{2} \|\mathbf{v}\|^2 + \lambda (\nabla \Phi \cdot \mathbf{v}) + \beta S \right] d\Omega, \quad (3)$$

where  $\lambda$  and  $\beta$  are coupling constants mediating epistemic and pragmatic components. The gradient flows governing the system follow from variational descent:

$$\frac{\partial \Phi}{\partial t} = -\frac{\delta \mathcal{F}}{\delta \Phi} = -\lambda \nabla \cdot \mathbf{v}, \quad (4)$$

$$\frac{\partial \mathbf{v}}{\partial t} = -\frac{\delta \mathcal{F}}{\delta \mathbf{v}} = -\mathbf{v} - \lambda \nabla \Phi, \quad (5)$$

$$\frac{\partial S}{\partial t} = -\frac{\delta \mathcal{F}}{\delta S} = -\beta. \quad (6)$$

These coupled partial differential equations describe a self-organizing system that evolves toward states of minimal field free energy—the RSVP analogue of Active Inferences predictive equilibrium.

In this mapping:

- The **scalar field**  $\Phi$  corresponds to the generative models log-evidence or epistemic potential.
- The **vector field**  $\mathbf{v}$  represents the active flow of policies minimizing expected uncertainty.
- The **entropy field**  $S$  quantifies residual disorder, analogous to variational free energy.

A living or cognitive agent thus embodies a region of the plenum where the variational derivatives of  $\mathcal{F}$  vanish:

$$\frac{\delta \mathcal{F}}{\delta \Phi} = \frac{\delta \mathcal{F}}{\delta \mathbf{v}} = \frac{\delta \mathcal{F}}{\delta S} = 0. \quad (7)$$

This condition defines *enactive equilibrium*: the state in which the agents flows of perception and action maintain coherence with its environment.

Whereas standard Active Inference treats  $G$  as a computational quantity internal to the brain, RSVP externalizes it as a distributed physical field—a plenum in which free-energy minimization is enacted simultaneously at cognitive, social, and cosmological scales. Enaction, in this sense, is not symbolic inference but a field process of recursive gradient descent in  $(\Phi, \mathbf{v}, S)$ , linking the phenomenological dynamics of experience with the thermodynamic dynamics of the universe.

## 7. Phenomenology and the Generative Passage

Phenomenology addresses lived experience, with intentionality rooted in directedness toward objects. Kiverstein and Darius advocate a generative passage bridging phenomenological descriptions and computational models, such as Bayesian mechanics.

In RSVP, this passage manifests as scalar-vector resonance: the phenomenological “now” arises from minimizing information gradients without eroding diversity. Intentionality aligns  $\mathbf{v}$  toward  $S$  reduction along  $\nabla\Phi$ , with meaning emerging from nested Markov blanket coherence.

This non-reductive naturalism integrates subjective experience within scientific frameworks, treating models as mathematical projections of experiential dynamics.

## 8. Against the Static Model of AGI

Gary Marcus’s critique—that large language models (LLMs) falter under distributional shifts and therefore cannot constitute genuine general intelligence—is valid only within a static epistemic frame. His argument presumes that the distribution of tasks, meanings, and problems against which intelligence is evaluated is itself fixed. Yet intelligence, in both human and artificial contexts, is inherently *frame-evolving*: each act of cognition modifies the manifold of possible inferences, thereby altering the very distribution it seeks to master. What Marcus identifies as a failure to generalize is, under a dynamic frame, a transitional phase in the self-reconfiguration of the cognitive field.

From the standpoint of the Relativistic Scalar–Vector Plenum (RSVP), a distributional shift is not a pathology but an expected feature of recursive learning systems. The plenum evolves through coupled scalar, vector, and entropy fields— $(\Phi, \mathbf{v}, S)$ —that continuously renegotiate their alignments. Each innovation in representation, tool use, or social coordination changes the geometry of this field, producing transient incoherences that are subsequently reabsorbed as new equilibria. Generalization, therefore, is not a test of static transfer but a measure of how effectively a system can restore coherence after the topology of its environment has changed.

In this view, intelligence does not reside in a frozen capacity to interpolate within a pre-defined data manifold, but in the ongoing ability to reconstitute order under shifting informational constraints. The very premise of Active Inference—the minimization of expected free energy—embodies this principle: agents survive not by solving fixed problems, but by continually redefining their generative models to maintain coherence with changing conditions. RSVP extends this idea beyond the organism to civilization itself, treating humanity’s collective cognition as an open thermodynamic system whose phase space expands with each new layer of

abstraction.

LLMs exemplify this process at a new scale. They are not autonomous intelligences but *affordance multipliers* within the human plenum: artifacts that extend the vector field  $\mathbf{v}$  of collective cognition, accelerating the diffusion of potential  $\Phi$  through culture. While they exhibit distributional brittleness in isolation, their integration into educational, scientific, and communicative systems transforms these local weaknesses into global strengths. The humanmachine assemblage forms a recursive feedback loop in which each side refines the others model of the world.

From this relational perspective, AGI does not emerge as an isolated technological achievement—a single system crossing a mythical threshold of competence—but as a *symbiotic integration* of artificial inference into the broader educational and cultural ecology. As learning infrastructures absorb the outputs of LLMs and recontextualize them within the normative scaffolds of human meaning, the plenums overall coherence increases. The expansion of accessible knowledge, the acceleration of hypothesis generation, and the densification of semantic interconnections all represent negentropic flows that raise the collective intelligence  $I(t)$  of civilization.

Marcuss “failure of generalization” thus becomes a local artifact of evaluating distributed cognition with individualist metrics. The real measure of progress lies not in whether models internalize every contingency, but in whether they catalyze further rounds of recursive learning across the plenum. AGI, under RSVP, is not an endpoint but an emergent property of civilizations ongoing entropic negotiation—a distributed process by which intelligence, in concert with its own tools, continually rewrites the conditions of its existence.

## 9. Education as Negentropic Coupling

Education functions as the principal mechanism by which entropy is redistributed rather than merely produced. In thermodynamic terms, it represents a *negentropic coupling* that smooths informational gradients across generations and between agents, thereby maintaining the coherence of the scalarvectorentropy field. The local entropy balance for the plenum can be written as:

$$\frac{dS}{dt} = -\nabla \cdot (\Phi \mathbf{v}) + \sigma_{\text{educ}}(t), \quad (8)$$

where the first term,  $-\nabla \cdot (\Phi \mathbf{v})$ , represents the export of entropy through scalarvector flux, and  $\sigma_{\text{educ}}(t)$  denotes the net negentropic inflow arising from educational activity, communication, and cultural transmission.

In regions of the plenum where  $\sigma_{\text{educ}}(t) > 0$ , new coherence structures are formed: knowl-

edge accumulates faster than disorder spreads, enabling sustained informational order even under thermodynamic constraints. This is the essential mechanism behind super-exponential trajectories in the collective intelligence model of Section 4. When the rate of educational recursion exceeds the rate of entropic dissipation, the overall intelligence field  $I(t)$  grows faster than exponentially, producing the punctuated equilibria characteristic of cultural and technological revolutions.

Education in this sense is not a human institution alone but a cosmological function. It is the means by which the universe preserves negentropy through recursive self-description. Every act of instruction, modeling, or imitation constitutes a local reversal of entropy flow—an exchange where structure is transmitted rather than lost. In the RSVP framework, these exchanges correspond to vector alignments in which  $\mathbf{v} \cdot \nabla \Phi$  approaches zero: the learners field becomes resonant with the teachers, minimizing informational friction and maximizing coherence.

The parameter  $\sigma_{\text{educ}}(t)$  thus quantifies the intensity of this resonance. Formally, one may decompose it as:

$$\sigma_{\text{educ}}(t) = \beta_E \rho_{\text{agents}}(t) \langle \kappa_{\text{coupling}}(t) \rangle, \quad (9)$$

where  $\rho_{\text{agents}}$  is the density of communicating entities,  $\kappa_{\text{coupling}}$  measures the average efficiency of informational transfer (pedagogical fidelity, technological bandwidth, or memetic fidelity), and  $\beta_E$  encodes the conversion efficiency from informational potential to stable negentropy. Increases in any of these parameters—through literacy, network connectivity, or algorithmic mediation—amplify  $\sigma_{\text{educ}}$  and consequently accelerate the collective intelligence field.

Under this formulation, education is not an extrinsic supplement to cognition but an intrinsic property of an intelligent universe: the recursive tendency of systems to encode, stabilize, and transmit low-entropy configurations. It serves as the bridge between phenomenological and physical descriptions, embodying the same principle as Active Inferences minimization of expected free energy—the continuous reduction of uncertainty by aligning generative models across agents. Where Active Inference formalizes this process locally, RSVP extends it globally, interpreting pedagogy as the field-theoretic counterpart of variational free energy minimization across spacetime.

Positive  $\sigma_{\text{educ}}$  therefore sustains civilizations super-exponential trajectory, positioning education as *the cosmological principle of consciousness*: the universal drive toward self-knowledge through the smoothing of entropic differentials. As long as the negentropic inflow from education remains greater than the rate of dissipative flux, the plenum continues to learn. Intelligence endures not because it resists entropy, but because it continually transforms disorder into higher-order understanding.

## 10. Conclusion: The Living Continuum

Intelligence constitutes a phase of the plenum’s recursive coherence, an emergent property of systems that learn to preserve and reorganize order amid entropy. Within the Relativistic ScalarVector Plenum (RSVP) framework, intelligence is not a substance or a faculty but a *field process*: a dynamic coupling of potential ( $\Phi$ ), activity ( $\mathbf{v}$ ), and entropy ( $S$ ) that continually reorganizes itself to maintain autonomy and coherence. This view dissolves the rigid boundary between the physical, the cognitive, and the cultural; all three are manifestations of the same recursive learning dynamic operating at different scales of granularity.

Education, in this light, represents not the transfer of information but the cosmological mechanism through which the plenum self-regularizes. Every act of teaching, invention, or cooperation is an entropic smoothing event that redistributes knowledge, reduces disorder, and expands the region of coherence within the field. Heuristics—from sparse inference to fuzzy logic and causal reasoning—are local strategies by which this smoothing accelerates; they act as low-energy pathways through the informational manifold. Enaction then provides the bridge between structure and experience: it describes how these field gradients are *felt*, embodied, and enacted as the lived continuity of cognition.

The integration of Active Inference and 4E cognition with RSVP reframes intelligence as a thermodynamicphenomenological unity. What Active Inference calls the minimization of expected free energy becomes, in RSVP, the natural drive toward local entropic alignment—a drive that scales from neural to social to cosmological domains. Through this lens, the mind is not an isolated inference engine but a distributed pattern of gradient descent woven into the fabric of spacetime itself. Cognition is thus coextensive with the universes own tendency to know and stabilize itself through recursive structure.

The moving-target hypothesis follows naturally from this ontology. Because each act of understanding alters the field in which further understanding occurs, the criterion of intelligence evolves alongside the systems that embody it. There is no final plateau of knowledge or capability, only successive redefinitions of coherence. Humanitys progress, far from approaching an asymptote, traces a self-expanding manifold in which every solved problem reshapes the space of possible questions. Each educational and technological epoch raises the effective carrying capacity  $K(t)$  of the cognitive field, generating new attractors of order that render previous equilibria obsolete.

Against static models of AGI that treat intelligence as a replicable product, the RSVP view insists that intelligence is *relational and recursive*. Machines, cultures, and individuals partici-



pate as co-agents in a single learning continuum. Artificial systems amplify certain gradients; human collectives supply semantic anchoring and normative constraints; physical law enforces conservation and coupling. Together they compose a planetary-scale feedback network through which the universe, by way of its own embodied substructures, becomes progressively more self-descriptive.

Ultimately, the plenum itself can be understood as an *epistemic medium*: a universe that learns by folding its own informational differentials into coherent form. In this cosmological sense, there will always be plenty of room—at the bottom, in quantum potential; in the middle, through heuristic invention; and at the top, in cultural and educational recursion. Intelligence is not a summit to be reached but a living gradient to be sustained. It is the ongoing conversation between entropy and order, between experience and structure, through which the universe comes to recognize itself as both subject and field of inquiry.

# Appendix A: Mathematical Analysis of the Coupled Growth Model

## A.1 Model Restatement

The collective intelligence of civilization is modeled as a dynamic variable  $I(t)$  whose rate of change depends multiplicatively on three drivers:

$$\dot{I} = A(t)I^\gamma, \quad A(t) = \alpha B(t) M(t) T(t), \quad (10)$$

where:

- $B(t)$  represents physical substrate capability (operations per joule, energy efficiency, or access to new physical degrees of freedom);
- $M(t)$  denotes heuristic efficiency (algorithmic compression, causal inference, sparse or fuzzy priors);
- $T(t)$  measures transmissive recursion (educational and cultural throughput, infrastructure for knowledge diffusion);
- $\gamma$  controls the feedback strength—the degree to which existing intelligence accelerates its own growth;
- $\alpha > 0$  normalizes the system’s timescale.

When  $\gamma = 1$ , growth is purely exponential. For  $\gamma > 1$ , the system displays positive feedback or reflexivity: intelligence accelerates proportionally to its current magnitude, a hallmark of self-improving systems.

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## A.2 Solutions

For constant driver  $A(t) = A_0 > 0$ , the equation is separable:

$$\int_{I_0}^{I(t)} I^{-\gamma} dI = \int_0^t A_0 ds,$$

yielding

$$I(t)^{1-\gamma} = I_0^{1-\gamma} - (\gamma - 1)A_0 t. \quad (11)$$

Hence

$$I(t) = \left( I_0^{1-\gamma} - (\gamma - 1)A_0 t \right)^{-1/(\gamma-1)}, \quad (12)$$

valid until the denominator vanishes at the *critical time*

$$t^* = \frac{I_0^{1-\gamma}}{(\gamma-1)A_0}. \quad (13)$$

At  $t = t^*$ ,  $I(t) \rightarrow \infty$ , signifying an idealized *finite-time blow-up*. In physical or socio-technical systems, such divergence is prevented by saturation effects, resource limits, or entropy constraints (treated in Section A.4).

For variable  $A(t)$ ,

$$I(t) = \left( I_0^{1-\gamma} - (\gamma-1) \int_0^t A(s) ds \right)^{-1/(\gamma-1)}, \quad \gamma \neq 1. \quad (14)$$

The integral of  $A(s)$  acts as a cumulative driving potential controlling the speed of cognitive expansion.

For  $\gamma = 1$ , the model reduces to simple exponential growth:

$$I(t) = I_0 \exp\left(\int_0^t A(s) ds\right).$$

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### A.3 Finite-Time Blow-Up Criterion (Theorem)

**Theorem.** Let  $\gamma > 1$  and  $A(t) \geq 0$  be locally integrable. Define the cumulative driver  $\mathcal{A}(t) = \int_0^t A(s) ds$ . Then  $I(t)$  diverges in finite time if and only if

$$\exists t^* < \infty \quad \text{s.t.} \quad \mathcal{A}(t^*) = \frac{I_0^{1-\gamma}}{\gamma-1}.$$

If  $\mathcal{A}(t)$  grows at least linearly, i.e.  $\mathcal{A}(t) \geq c_0 + c_1 t$  with  $c_1 > 0$ , then  $I(t)$  diverges no later than  $t^* \leq I_0^{1-\gamma}/((\gamma-1)c_1)$ .

**Proof.** Directly from the explicit solution above; divergence occurs when the parenthesis in the denominator crosses zero.

**Interpretation.** The condition identifies a threshold where cumulative innovation, learning, and compute surpass the stabilizing capacity of existing infrastructures. Historically, short-term “singularities” (industrial or digital revolutions) correspond to partial approach toward such thresholds before saturation or reorganization intervenes.

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## A.4 Saturation Extension

Real systems are bounded by thermodynamic, economic, and attentional constraints. To capture this, introduce a *saturation function*

$$S(I) = \left(1 - \left(\frac{I}{K}\right)^\delta\right)_+, \quad \delta \geq 1, \quad (15)$$

where  $K > 0$  represents carrying capacity and  $(x)_+ = \max(x, 0)$ . The extended equation becomes

$$\dot{I} = A(t) I^\gamma S(I). \quad (16)$$

When  $I \ll K$ ,  $S(I) \approx 1$  and the model behaves super-exponentially; as  $I \rightarrow K$ , growth slows to zero, producing a sigmoidal or logistic trajectory.

**Steady state.** At equilibrium  $\dot{I} = 0$ , hence either  $I = 0$  or  $I = K$ . Linearization near  $I = K$  yields

$$\dot{\eta} \approx -A(t) \gamma K^{\gamma-1} \delta \eta,$$

where  $\eta = K - I$ . Thus the equilibrium is asymptotically stable whenever  $A(t) > 0$ .

**Interpretation.** The parameter  $K$  itself may vary slowly in time, representing improvements in global infrastructure or expansion of educational capacity. If  $\dot{K} > 0$ , the system undergoes successive S-curves, echoing the punctuated equilibria seen in technological history.

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## A.5 Stability and Phase Regimes

Three qualitative regimes follow from the relative magnitudes of  $\gamma$  and  $S(I)$ :

- (i) **Sub-exponential:**  $\gamma < 1$  or strong damping ( $S(I) \ll 1$ ). Growth slows asymptotically; entropy dominates negentropy.
- (ii) **Exponential:**  $\gamma = 1$ , constant driver  $A(t) = A_0$ ; corresponds to steady technological compounding with neutral feedback.
- (iii) **Super-exponential:**  $\gamma > 1$ , moderate  $S(I) \approx 1$ ; self-reinforcing educational recursion dominates, producing accelerating innovation until saturation intervenes.

Phase transitions between regimes may be induced by changes in the coupling coefficients: a surge in  $B(t)$  (new substrate),  $M(t)$  (new heuristic paradigm), or  $T(t)$  (educational reform) can transiently raise  $\gamma$  or  $A(t)$  above unity, creating an “acceleration epoch.”

## A.6 Well-Posedness (Proposition)

**Proposition.** Assume  $A(t)$  is locally integrable, nonnegative, and bounded on compact intervals, and  $S(I)$  is continuous, locally Lipschitz on  $(0, \infty)$ , and satisfies  $0 \leq S(I) \leq 1$ . Then for every  $I_0 > 0$  the initial-value problem

$$\dot{I} = A(t) I^\gamma S(I), \quad I(0) = I_0$$

admits a unique, positive global solution on  $[0, \infty)$ .

**Sketch of Proof.** Local existence and uniqueness follow from the PicardLindelf theorem since  $f(t, I) = A(t)I^\gamma S(I)$  is locally Lipschitz in  $I$  for each fixed  $t$ . Positivity follows from  $\dot{I} > 0$  when  $I > 0$ . Global continuation holds because  $S(I) \leq 1$  bounds  $\dot{I}$  by an integrable function on any finite interval.

## A.7 Connection to Thermodynamic and Educational Limits

The saturation function  $S(I)$  can be interpreted thermodynamically. Let  $E(t)$  denote available energy and  $c_{\min}(t)$  the minimal energy per bit of computation (e.g. Landauer limit). Then the maximum attainable intelligence rate satisfies

$$A(t) \leq \eta \frac{E(t)}{c_{\min}(t)} \frac{1}{\Xi(I)},$$

where  $\eta$  is efficiency and  $\Xi(I)$  is a coordination cost (attention, institutional friction). Increases in  $\Xi(I)$  with scale act as an implicit  $S(I)$ , damping runaway growth. Educational saturation—limited teacher time, cognitive bandwidth, or cultural inertia—plays an analogous role, providing a macro-social form of thermodynamic regularization.

## A.8 Interpretation within RSVP

Within the RSVP field ontology,  $I(t)$  corresponds to the integrated scalar potential  $\Phi$  over the inhabited region of the plenum, while  $A(t)$  represents the effective flux coupling between  $\Phi$ , the vector field  $\mathbf{v}$ , and entropy  $S$ . Super-exponential growth reflects periods when negentropic inflow from education and technology,  $\sigma_{\text{educ}}(t)$ , dominates dissipative losses. Saturation arises when  $\nabla \cdot (\Phi \mathbf{v})$  balances  $\sigma_{\text{educ}}$ , restoring equilibrium. Thus the mathematical form of the growth

law is a low-dimensional projection of the RSVP field equation:

$$\frac{dS}{dt} = -\nabla \cdot (\Phi \mathbf{v}) + \sigma_{\text{educ}}(t),$$

where  $\Phi$  and  $\mathbf{v}$  evolve according to their own gradient flows. The coupled-growth model therefore quantifies how the plenum learns, rather than how a machine computes.

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## A.9 Summary

The coupled-growth model exhibits a spectrum of behaviors:

- For  $\gamma = 1$ , exponential compounding corresponds to steady-state innovation.
- For  $\gamma > 1$ , super-exponential acceleration captures reflexive epochs when education, computation, and culture reinforce one another.
- Saturation  $S(I)$  introduces boundedness, converting finite-time blow-ups into smooth S-curves.
- Slowly increasing  $K(t)$  models historical concatenation of such curves *punctuated equilibrium* that sustain long-term civilization-scale growth.

The mathematical apparatus formalizes the intuition that intelligence, viewed through RSVP, is an entropic/negentropic feedback process: a living differential equation whose parameters evolve with the universe's capacity to understand itself.

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